Max-Planck-Institut für Radioastronomie

Timing the Double Neutron Star PSR J1756−2251

Two decades of timing baseline, some questions answered, more questions asked.

Fazal Kareem, P. C. C. Freire, V. Venkatraman Krishnan · Max Planck Institute for Radio Astronomy, Bonn · on behalf of the MeerTIME / RelBin collaboration

1 A laboratory for strong-field gravity

PSR J1756−2251 is a recycled pulsar in a relativistic double neutron star system with a 7.67-h, mildly eccentric orbit. We extend its timing baseline by six years through dense MeerKAT L/S-band and Parkes UWL observations, bringing the total span to over two decades. The extended data set yields improved measurements of five post-Keplerian parameters, tightening constraints on the component masses and the system's relativistic dynamics.

28.46 ms SPIN PERIOD	7.67 h ORBITAL PERIOD	0.1805 ECCENTRICITY
1.34 M☉ PULSAR MASS	1.23 M☉ COMPANION MASS	2004–26 TIMING BASELINE

2 Two decades, 7 telescopes, 12 Backends

Telescope	Backend	Frequency (GHz)	ToAs	Date span
Legacy (Ferdman et al. 2014): Parkes, Green Bank, Nançay, Jodrell Bank, Westerbork	Various	1.4–1.5	8 775	2003–2013
Parkes	UWL / Medusa	0.7–4.0	22 906	2020–2023
MeerKAT	L-band (775 / 856 MHz BW)	0.9–1.7	5 840	2019–2022
MeerKAT	S-band (875 MHz BW)	2.0–2.8	1 624	2023–2025
Total	**12**	**0.7-4.0**	**39 145**	**2003–2025**

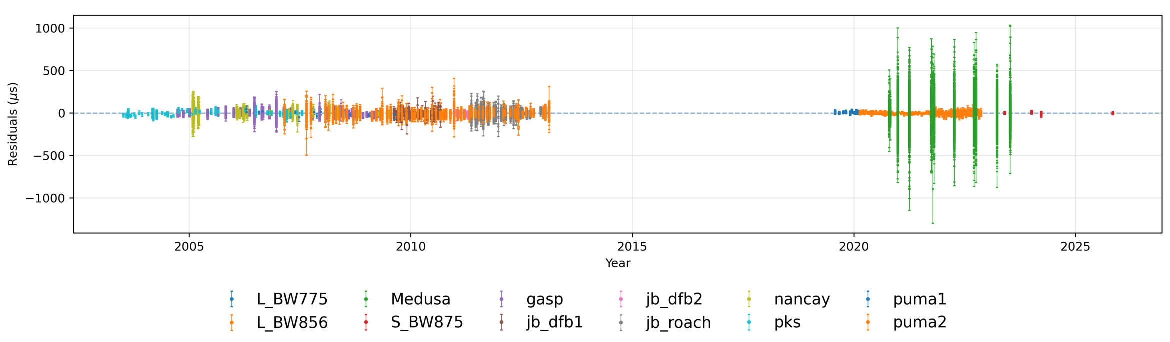


Fig. 1 *Timing residuals from 12 backends under the DDH timing model with the favoured white-noise, red-noise, and DM noise model.*

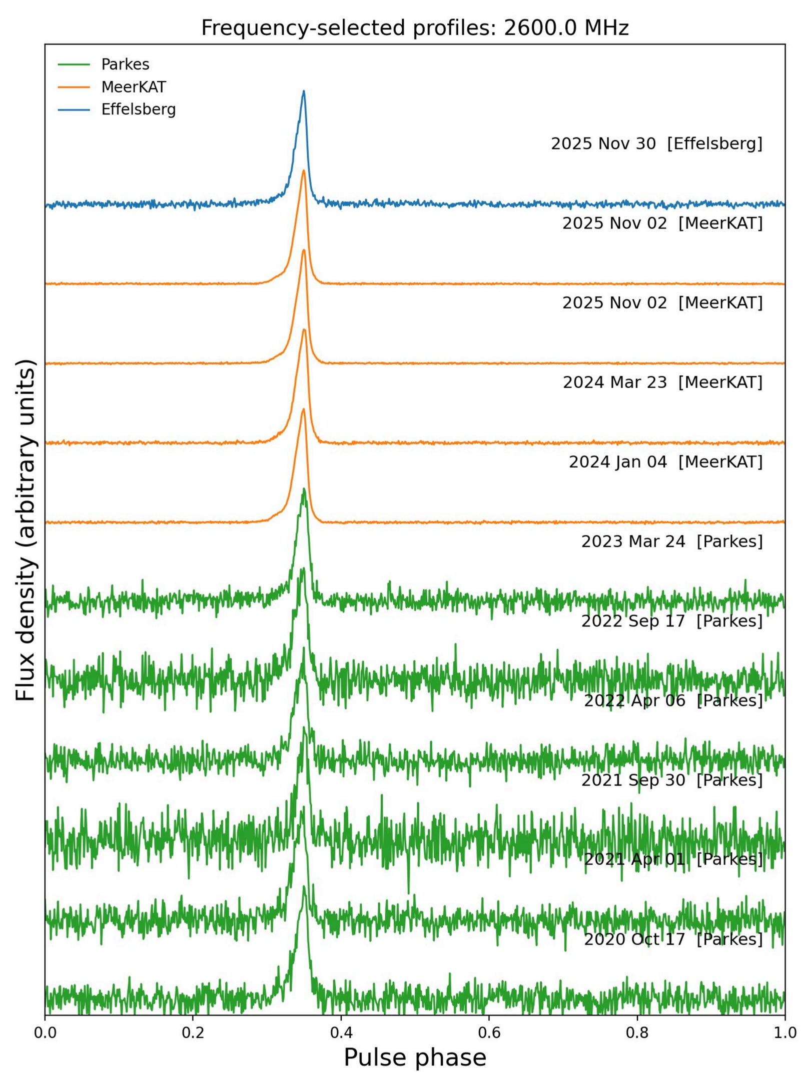


Fig. 2 *Profiles matched at 2.6 GHz with a bandwidth of 200MHz from Parkes (green), MeerKAT (orange), and Effelsberg (blue) spanning 2020–2025, showing a consistent, sharp main pulse across all three telescopes.*

3 Better constraints on everything!

Parameter	This work (DDH/DDGR)	Ferdman et al. (2014)
RAJ (hms)	17:56:46.632944(47)	17:56:46.633812(15)
DECJ (dms)	−22:51:59.419(57)	−22:51:59.35(2)
PMRA (mas/yr)	−2.015(79)	−2.42(8)
PMDEC (mas/yr)	9.9(7.0)	<20
PX (mas)	−2.02(1.66)	1.05(55)
F0 (Hz)	35.135072515400(17)	35.135072714 5469(6)
F1 (s^{-2})	$-1.255741(95)\times10^{-15}$	$-1.256079(3)\times10^{-15}$
DM (cm^{-3} pc)	121.158(39)	121.196(5)
PB (d)	0.31963390155609(34)	0.31963390143(3)
A1 (lt-s)	2.7564799(27)	2.756457(9)
ECC	0.18056971(11)	0.1805694(2)
OM (deg)	322.52357(10)	327.8245(3)
OMDOT (deg/yr)	2.5823837(50)	2.58240(4)
GAMMA (ms)	1.1317(27)	1.148(9)
PBDOT (s/s)	$-2.02693(25)\times10^{-13}$	$-2.29(5)\times10^{-13}$
Shapiro r (M☉)	—	1.6(6)
Shapiro s = sin i	—	0.93(4)
H3 (s)	$1.611(44)\times10^{-6}$	—
STIG	0.687(21)	—
XPBDOT (s/s)	$1.3955(25)\times10^{-14}$	$1.8(6,9)\times10^{-14}$
XDOT (lt-s s)	$2.321(677)\times10^{-15}$	—
M2 (M☉)	1.2300(20)	1.230(7)
Mp (M☉)	1.3400(20)	1.341(7)

Improved measurements of five post-Keplerian parameters.

- The periastron advance $\dot{\omega}$ and Einstein delay γ are improved by factors of 38 and 22, respectively.
- The orbital period derivative $\dot{P}_b$ is now measured at a fractional precision of 0.012%.
- The orthometric Shapiro-delay amplitude h_3 and ratio ς are measured here for the first time, providing two independent Shapiro-delay constraints.

Together, these improvements tighten the pulsar and companion mass estimates by a factor of ~9.

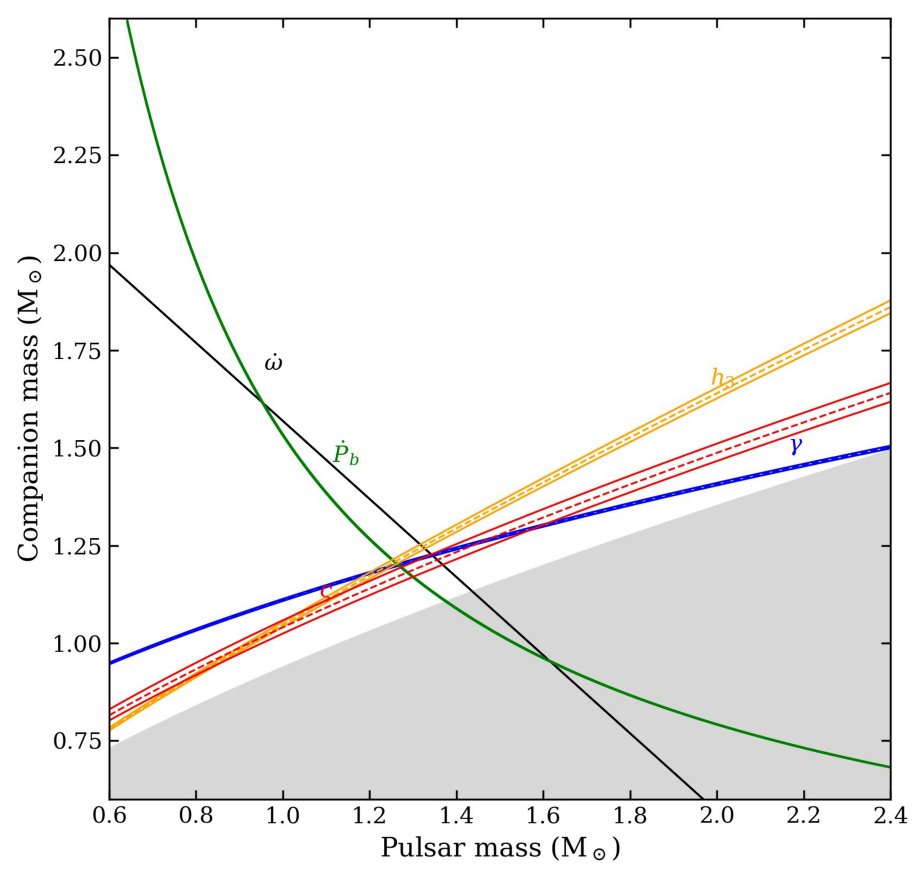


Fig. 4 *Updated mass–mass diagram from the DDGR + red-noise solution, showing five post-Keplerian constraints ($\dot{\omega}$, γ, $\dot{P}_b$, and the orthometric Shapiro-delay parameters) with ±1σ bands; the grey region is excluded by sin i ≤ 1.*

In earlier studies, one could easily see the effect of a biased $\dot{P}_b$. This observed departure from the GR prediction can be attributed to the combined effects on the observed orbital decay of this system that are resulting from kinematic biases. The robustness of any gravity test using the $\dot{P}_b$ we derive from timing this pulsar is limited until we are able to better constrain the systematic effects that influence its measurement.

We have not been able to constrain these systematics in the latest timing analysis. We note a high uncertainty in the proper motion of the pulsar and an unconstrained parallax.

4 Orbital decay and distance

The DDGR timing model can be reparametrised to yield the excess orbital-period derivative, $\Delta\dot{P}_b \equiv \dot{P}_{b,obs} - \dot{P}_{b,GR}$, which captures the kinematic contribution from proper motion (the Shklovskii effect) and differential Galactic acceleration along the line of sight. Under the assumption that the intrinsic orbital decay is entirely due to gravitational-wave emission as predicted by general relativity, we invert the kinematic bias equation to solve for the pulsar distance.

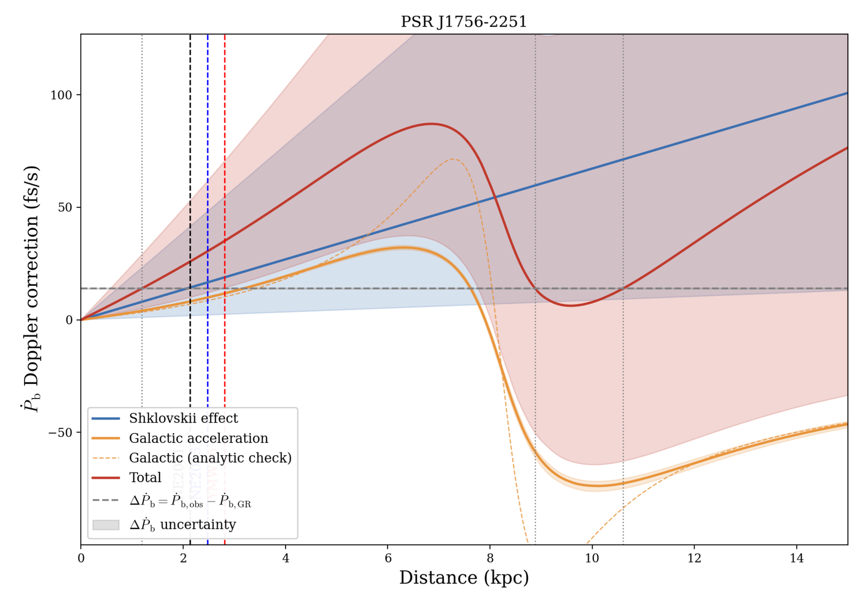
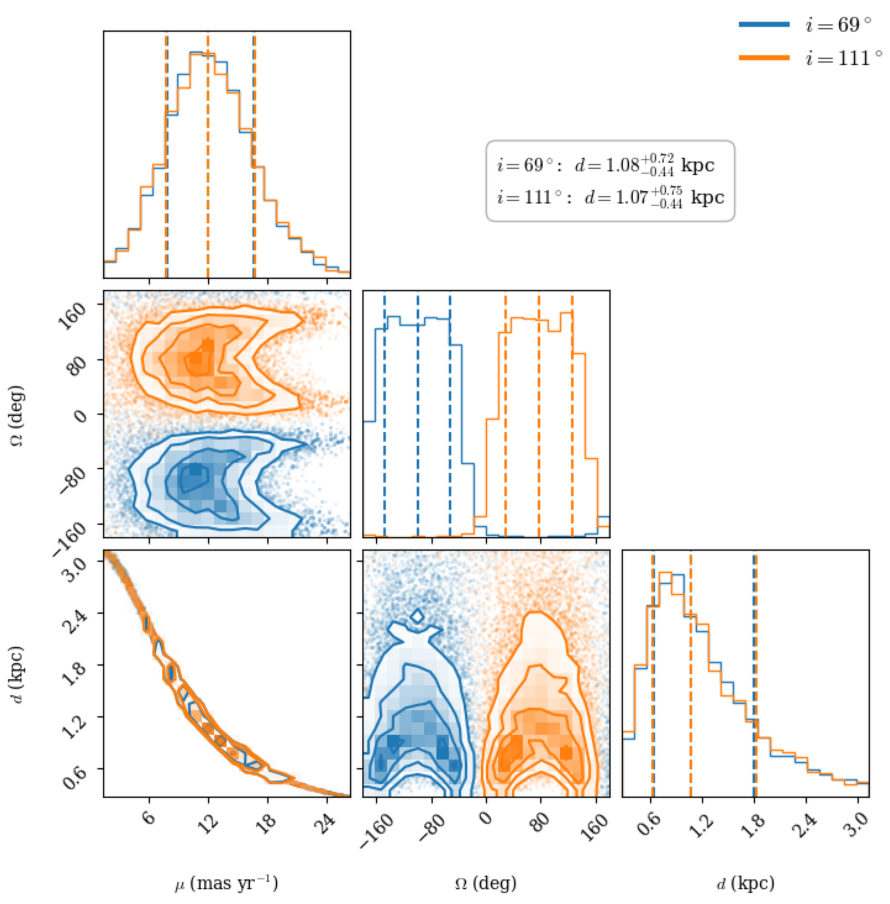


Fig. 5 *Kinematic $\dot{P}_b$ corrections as a function of distance. The Shklovskii and Galactic acceleration terms sum to the measured excess $\Delta\dot{P}_b$ (dashed line), yielding a distance estimate; DM-model distances from NE2001, NE2025, and YMW16 are indicated.*

We use $\dot{x}$ (Kopeikin proper-motion term, marginalized over the node Ω) to constrain μ, then convert to distance via the orbital-decay excess, with the Galactic term from galpy (MWPotential2014). A Monte Carlo over μ, Ω, R_0/V_0, and XPBDOT, weighted by the $\dot{x}$ likelihood, gives posteriors on d.



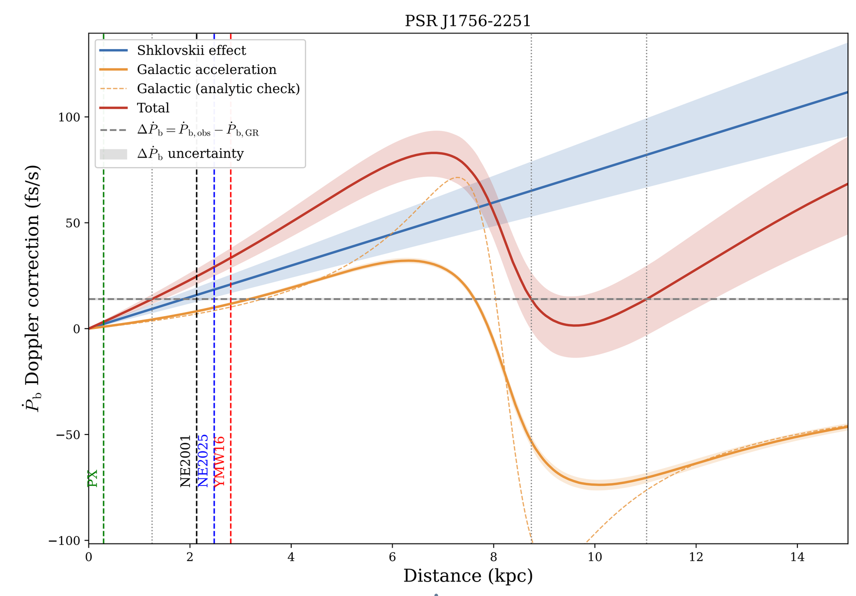


Fig. 6 a, b *Updated kinetic $\dot{P}_b$ corrections using the $\dot{x}$ posterior.*

This is a preliminary result, we are still analysing this method.

5 Pulsar position and solar wind

The proximity of PSR J1756−2251 to the ecliptic makes the timing sensitive to solar wind dispersion. In this analysis, times of arrival obtained during closest solar approach have been excised; a dedicated solar wind noise model will be incorporated in future iterations.

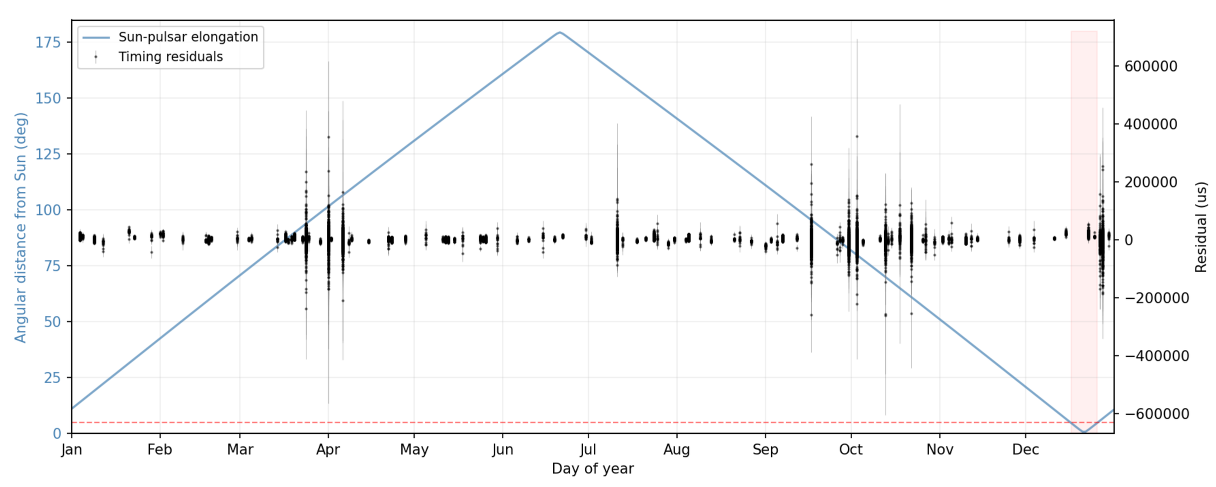


Fig. 7 *Annual angular separation of PSR J1756−2251 from the Sun, overlaid on the timing residuals. ToAs obtained within 5° of solar conjunction, where solar wind dispersion is significant, are highlighted.*

6 Seven years of polarimetric analysis with MeerKAT

Every MeerKAT epoch is examined for polarisation variability. Stacked L and V profile variations, differenced Stokes Q/U images and position-angle maps shows interesting changes that are not seen in the pulse intensity profile. The cause of this is still under investigation and any suggestions are welcome!

Fig. 8 *Faraday-corrected position-angle swings colour-coded by MJD, revealing a persistent orthogonal mode jump whose phase shows a marginal drift over seven years.*

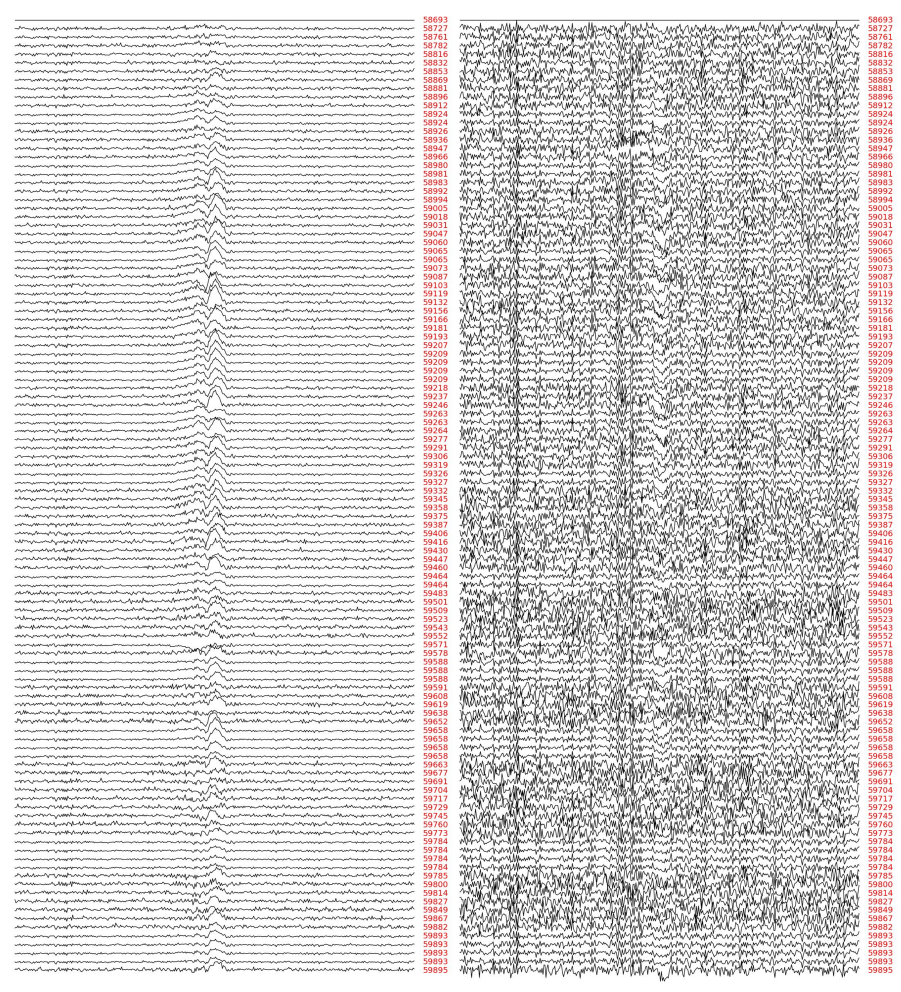


Fig. 9 *Variance in linear (L) and circular (V) polarisation with respect to the first epoch for MeerKAT campaign*

7 Gravity passes and the test goes on.

- Every measured post-Keplerian parameter is consistent with general relativity at current precision.
- The kinematic distance from $\Delta\dot{P}_b$ is consistent with all DM-based electron-density models; the timing parallax remains unconstrained.
- Polarimetric variability - absent in total intensity raises open questions about the emission geometry.
- Continued Effelsberg monitoring will tighten the component masses and kinematic $\dot{P}_b$ correction year on year.
- VLBI astrometry is needed to independently confirm the proper motion and resolve the distance ambiguity.

KEY REFERENCES

[1] Faulkner et al. 2005, ApJ 618, L119
[2] Ferdman et al. 2014, MNRAS 443, 2183
[3] Damour & Deruelle 1986
[4] Hobbs et al. 2006, MNRAS 369, 655
[5] S. M. Kopeikin 1996 *ApJ* 467 L93

Poster and References

MeerKAT is operated by SARAO; Murriyang (Parkes) by CSIRO; the Effelsberg 100-m telescope by the MPIfR.

fkareem@mpifr-bonn.mpg.de | fazalkareem.com